Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 2 Quiz 1 - Kinematics in Two Dimensions

Mock Exam F · Form 1: Advanced reasoning

Multiple Choice Questions: Select the best answer. (1 point each)

1. Two balls are launched simultaneously in one vertical plane with different velocity vectors. Both remain in free flight. What shape does the relative-position vector trace?

(A) Always a parabola

(B) Circle

(C) Stationary point

(D) Straight line

2. A student resolves a 20 m/s velocity directed 30° north of west. East is +x and north is +y. What is its y component?

(A) -10 m/s

(B) 20 m/s

(C) 5 m/s

(D) 10 m/s

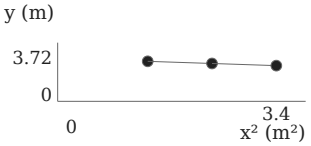
3. A ball leaves a 4 m ledge horizontally at 5 m/s. Use g=10 m/s² and consider airborne frames only. x² (m²) -> y (m): (1.4, 3.72); (2.4, 3.52); (3.4, 3.32). If sensor gain and physical slope are both unknown, what breaks their ambiguity?

(A) Reverse the ordering of the data.

(B) Assume any straight graph proves exact calibration.

(C) Calibrate the sensor at two independently known reference values.

(D) Repeat the same unknown setting.



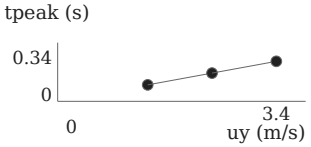
4. In gravity g=10 m/s² vary the upward component of launch velocity and time the ascent. Double gravitational acceleration. Plot the difference Ynew-Yold versus X. What slope should be observed?

(A) 0.04

(B) -0.05

(C) -0.02

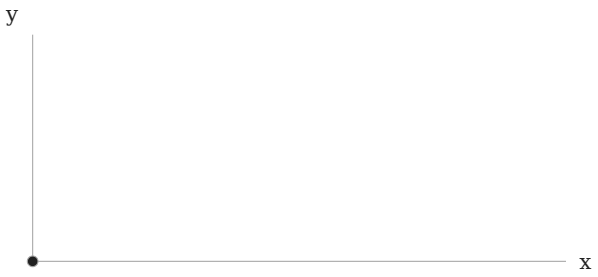
(D) -0.11



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A ball is launched from ground at 27 m/s, 45° above horizontal, and lands on the same ground. Ignore drag.

a. Draw the initial velocity vector and its horizontal and vertical components. **(1 pt)**



b. Find flight time, range, and velocity and acceleration at the top. **(3 pt)**

6. A projectile moves horizontally at +6 m/s while a target travels at +2 m/s. Track their horizontal separation during flight. t (s) -> xt-xp (m): (0.7, 9.2); (1.2, 7.2); (1.7, 5.2).

a. Build the model for moving target from physics and predict both slope and intercept. **(2 pt)**

b. Hold the target at its initial position. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A ball is launched from (0,h) at angle θ to hit (R,0), with $R > 0$ and $\cos\theta > 0$. Derive the required launch speed u and the condition for a positive solution. **(3 pt)**